

LOGARITHMIC RESIDUAL-COMPLEXITY BARRIERS FOR LOCAL MÖBIUS CANCELLATION

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ABSTRACT. Consider cancellation certificates for the Möbius function that pair square-free integers after removing their greatest common divisor. If the residual cofactors have at most $R(T)$ prime factors at scale $T = \max(a, b)$, every edge has length $O(T^\alpha)$, and only $O(X^\beta)$ vertices through X remain unmatched, then

$$\limsup_{X \rightarrow \infty} \frac{R(X) \log \log X}{\log X} \geq (1 - \alpha)(1 - \beta).$$

The square-root scale therefore forces a limsup of at least $1/4$. Finite vertex-disjoint, unit-weight, zero-sum Möbius cells with a common core compress exactly to opposite-sign pairs. When the residual budget depends on scale, the compression loses only endpoints near jumps of R . Thus short local certificates require logarithmically growing residual complexity, and common-core cells obey the same pairwise limitation. The results construct no matching and give no estimate for the actual Mertens function.

1. INTRODUCTION

Let μ be the Möbius function and

$$M(X) = \sum_{n \leq X} \mu(n).$$

The Riemann hypothesis is equivalent to $M(X) = O_\varepsilon(X^{1/2+\varepsilon})$ for every $\varepsilon > 0$; see Titchmarsh [7, Chapter XIV] or Montgomery–Vaughan [6, Chapter 15]. Since $\mu(n) \in \{-1, 0, 1\}$, a combinatorial route would pair most squarefree integers of opposite Möbius sign with short edges and charge the summatory error to unmatched vertices and edges crossing the cutoff.

We study the necessary nonlocality of a broad gcd-residual matching and a natural enlargement from pairs to finite cells. Related work treats sign patterns of μ and the Liouville function by analytic methods [4], together with harmonic and matrix structures on the squarefree integers [2]. McNew and Pomerance study coprime matchings between the divisor set of an integer and an initial interval, proving in particular that every squarefree integer is matchable [5]. Their edge condition concerns coprimality. The edges here must have opposite Möbius signs, short displacement, and a bounded residual after the common divisor is removed. The controlling question is the obstruction forced by the retained greatest common divisor in such an edge.

Aymone and Liu study analytic bounds for summatory functions of multiplicative functions resembling Möbius, including a separate one-quarter lower-bound exponent in Liu’s class [1, 3]. Their objects are altered multiplicative functions; the objects here are local matching certificates for the Möbius function itself.

Products of many primes in a short dyadic band form a large same-sign family. A small residual budget forces any repair edge incident to that family to retain a greatest

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common divisor larger than the permitted displacement. This gives the product threshold $(1 - \alpha)(1 - \beta)$ in Theorem 3.1.

The second part tests a larger zero-sum cell. Distinct squarefree vertices with unit Möbius weights and one common greatest common divisor can be rank-paired by sign. The pairing preserves the exact signed boundary mass at every cutoff. If the budget changes with the endpoint, failed derived pairs are confined to neighborhoods of its jump set; Theorem 5.2 quantifies the loss.

The scope is the stated certificate class. Weighted flows, overlapping cells, non-common-core structures, cross terms, signed residual estimates, and dense-jump budgets require separate arguments. The conclusions give no lower bound for $|M(X)|$ and do not cover every combinatorial approach to Möbius cancellation.

2. THE MAXIMAL RESIDUAL PAIR CLASS

Write $\omega(n)$ for the number of distinct prime divisors of n . Let $R : [1, \infty) \rightarrow \mathbb{N} \cup \{0\}$ be nondecreasing.

Definition 2.1 (Residual-admissible pair). Distinct squarefree integers a, b form an R -admissible pair if, with

$$g = (a, b), \quad T = \max(a, b),$$

one has

$$(1) \quad \omega(a/g) \leq R(T), \quad \omega(b/g) \leq R(T),$$

and $\omega(a/g)$ and $\omega(b/g)$ have opposite parity.

The parity condition gives $\mu(a) = -\mu(b)$, since $g, a/g, b/g$ are squarefree. Every such pair also obeys

$$(2) \quad |a - b| = g |a/g - b/g| \geq g.$$

Definition 2.1 permits every edge satisfying the stated budget, so any extra scheduling or graph restriction can only remove edges.

Lemma 2.2 (Matching identity). *Let $\mathcal{P}$ be a finite family of disjoint opposite-sign pairs and let $U(X)$ be the number of squarefree integers at most X not covered by $\mathcal{P}$. If $C(X)$ is the number of pairs with exactly one endpoint at most X , then*

$$|M(X)| \leq U(X) + C(X).$$

Proof. Pairs lying completely below or completely above X contribute zero. Each crossing pair and each uncovered squarefree vertex contributes a number of absolute value one. Integers with $\mu(n) = 0$ contribute nothing. $\square$

3. THE LOGARITHMIC COMPLEXITY THRESHOLD

Theorem 3.1 (Residual-complexity barrier). *Fix $0 \leq \alpha < 1$, $0 \leq \beta < 1$, and $C > 0$. Suppose a matching of squarefree integers uses only R -admissible pairs and satisfies*

$$(3) \quad |a - b| \leq C \max(a, b)^\alpha$$

on every used edge. If the number of unmatched squarefree vertices through X is $O(X^\beta)$, then

$$(4) \quad \limsup_{X \rightarrow \infty} \frac{R(X) \log \log X}{\log X} \geq (1 - \alpha)(1 - \beta).$$

Proof. Assume that the limsup in (4) is

$$r < (1 - \alpha)(1 - \beta).$$

Choose

$$(5) \quad \frac{r}{1 - \alpha} < \delta < 1 - \beta.$$

For large y , put $k = \lfloor y^\delta \rfloor$, let $m_y = \pi(2y) - \pi(y)$, let S_y contain the products of k distinct primes in $(y, 2y]$, and put $X_y = (2y)^k$.

The prime number theorem in a fixed dyadic interval [6] gives $m_y \gg y/\log y$. Since $k = y^{\delta+o(1)} = o(m_y)$, the standard upper and lower binomial estimates give

$$(6) \quad |S_y| = \binom{m_y}{k} = X_y^{1-\delta+o(1)}.$$

Every element of S_y has Möbius sign $(-1)^k$. The choice of δ in (5) makes this family larger than X_y^β by a fixed power for all sufficiently large y .

Fix $n \in S_y$ and suppose that it has a partner b . If $b \geq 2n$, then (3) gives $b^{1-\alpha} \leq 2C$, which fails for large y . Thus $b < 2n \leq 2X_y$, and

$$(7) \quad \frac{\log(2X_y)}{\log \log(2X_y)} = \left(\frac{1}{\delta} + o(1) \right) k.$$

The assumed limsup bound, monotonicity of R , and (5) yield some $\eta > 0$ such that

$$(8) \quad R(\max(n, b)) \leq R(2X_y) < (1 - \alpha - \eta)k$$

for all sufficiently large y .

Let $g = (n, b)$. The residual n/g contains at most $R(2X_y)$ of the k prime factors of n . Combining the retained prime factors with (2) and the short-edge bound gives

$$(9) \quad y^{(\alpha+\eta)k} < y^{k-R(2X_y)} \leq g \leq |n - b| \leq C(2X_y)^\alpha = y^{(\alpha+o(1))k},$$

which is impossible for large y . Hence every member of S_y is unmatched. Equation (6) contradicts the asserted $O(X_y^\beta)$ unmatched count. $\square$

Corollary 3.2 (Square-root matching scale). *Fix R . Suppose that for every $\varepsilon > 0$ there is a residual-admissible matching, allowed to depend on ε , whose edge lengths and unmatched counts are*

$$O_\varepsilon(X^{1/2+\varepsilon}).$$

Then necessarily

$$\limsup_{X \rightarrow \infty} \frac{R(X) \log \log X}{\log X} \geq \frac{1}{4}.$$

Proof. Apply Theorem 3.1 with $\alpha = \beta = 1/2 + \varepsilon$ and let $\varepsilon \downarrow 0$. $\square$

A matching construction additionally requires Hall-type expansion or another mechanism for controlling collisions.

4. COMMON-CORE CELLS COLLAPSE TO PAIRS

Definition 4.1 (Common-core Möbius cell). A cell is a finite set H of distinct squarefree positive integers with

$$\sum_{n \in H} \mu(n) = 0.$$

Write $g_H = \gcd\{n : n \in H\}$, $T_H = \max H$, and

$$S_H(X) = \sum_{\substack{n \in H \\ n \leq X}} \mu(n).$$

Theorem 4.2 (Exact fixed-budget pair compression). *Suppose a common-core Möbius cell H satisfies*

$$\omega(n/g_H) \leq R_H \quad (n \in H).$$

List its positive vertices increasingly as $p_1, \dots, p_m$ and its negative vertices increasingly as $q_1, \dots, q_m$. Pair p_i with q_i . Then every derived pair $\{a, b\}$ satisfies

$$\mu(a) = -\mu(b), \quad \omega(a/(a, b)), \omega(b/(a, b)) \leq R_H, \quad |a - b| \leq \text{diam}(H).$$

For every real X , exactly $|S_H(X)|$ of these pairs cross X .

Proof. The zero-sum hypothesis gives the same number of positive and negative vertices. For a paired a, b , the common core g_H divides $h = (a, b)$. Since the integers are squarefree,

$$\omega(a/h) \leq \omega(a/g_H) \leq R_H, \quad \omega(b/h) \leq \omega(b/g_H) \leq R_H.$$

Opposite Möbius signs give opposite residual parities. The diameter bound is immediate.

At a cutoff X , let $P(X)$ and $Q(X)$ count the positive and negative endpoints through X . Increasing rank pairing matches the common initial portion. The crossing-pair count is therefore $|P(X) - Q(X)| = |S_H(X)|$. $\square$

The final identity preserves the exact signed prefix mass simultaneously at every cutoff.

5. ENDPOINT BUDGETS AND SPARSE JUMPS

If a cell is admitted at budget $R(T_H)$, a derived pair with maximum endpoint $m < T_H$ may exceed the smaller budget $R(m)$. The following example shows the endpoint issue exactly.

Proposition 5.1 (Sharp finite endpoint gap). *Let*

$$H = \{2, 5, 21, 33\}, \quad R(t) = \begin{cases} 1, & t < 30, \\ 2, & t \geq 30. \end{cases}$$

Then H is a zero-sum common-core cell admitted at scale 33, but neither opposite-sign pair partition makes every pair admissible at its own maximum endpoint.

Proof. The signs are $(-1, -1, +1, +1)$ and every vertex has at most two prime factors. There are two opposite-sign pair partitions. In each partition, the pair not containing 33 has maximum below 30 and contains an endpoint with two residual prime factors, so it violates budget one. $\square$

Let $J(Y)$ be the number of integer jump locations $j \leq Y$ with $R(j) > R(j - 1)$.

Theorem 5.2 (Sparse-jump repair). *Let $R : \mathbb{N} \rightarrow \mathbb{N} \cup \{0\}$ and $D : \mathbb{N} \rightarrow [0, \infty)$ be nondecreasing. At a cutoff X , let $\mathcal{F}_X$ be a finite vertex-disjoint family of common-core Möbius cells satisfying*

$$\omega(n/g_H) \leq R(T_H), \quad \text{diam}(H) \leq D(T_H).$$

Rank-pair every cell as in Theorem 4.2, and delete a derived pair when it violates the residual budget at its own maximum endpoint. If $D(t) \leq t/2$ for all sufficiently large t , then, up to a fixed initial constant B_0 , the number $B(X)$ of deleted endpoints at most X obeys

$$(10) \quad B(X) \leq B_0 + (2\lfloor D(2X) \rfloor + 1)J(2X).$$

If also $D(t) \leq Ct^\alpha$ with $\alpha < 1$, every retained pair satisfies

$$(11) \quad |a - b| \leq C2^\alpha \max(a, b)^\alpha$$

for all sufficiently large endpoints.

Proof. Let a deleted pair have maximum endpoint m . Common-core inheritance still bounds both residuals by $R(T_H)$; deletion therefore implies $R(m) < R(T_H)$. Hence there is a jump $j \in (m, T_H]$. Every endpoint of the pair lies within $D(T_H)$ of j .

If a deleted endpoint $n \leq X$ lies in H , then

$$T_H - n \leq D(T_H) \leq T_H/2,$$

so $T_H \leq 2X$ and $D(T_H) \leq D(2X)$. Each endpoint is therefore an integer in a radius- $D(2X)$ neighborhood of a jump through $2X$. Vertex-disjointness ensures that no integer endpoint is counted twice, which proves (10).

For the retained-pair gap, put $m = \max(a, b)$. Eventually $m \geq T_H - D(T_H) \geq T_H/2$, and therefore

$$|a - b| \leq D(T_H) \leq CT_H^\alpha \leq C2^\alpha m^\alpha.$$

□

Corollary 5.3 (Compression at square-root scale). *Suppose $R(Y) = O(\log Y / \log \log Y)$ and, for every $\delta > 0$ and sufficiently large X , there is a family $\mathcal{F}_{\delta, X}$ as in Theorem 5.2 with*

$$D(t) = O_\delta(t^{1/2+\delta})$$

and total uncovered plus signed cell-boundary mass

$$O_\delta(X^{1/2+\delta}).$$

Here the boundary mass is the sum of $|S_H(X)|$ over the cells. Then for every $\varepsilon > 0$, that cutoff-specific cell certificate compresses to an opposite-sign pair certificate with unmatched plus crossing contribution $O_\varepsilon(X^{1/2+\varepsilon})$.

Proof. An integer-valued nondecreasing R has $J(Y) \leq R(Y) + O(1)$. Choose $0 < \delta < \min(\varepsilon, 1/2)$. Equation (10) then contributes

$$O_\delta\left(X^{1/2+\delta} \frac{\log X}{\log \log X}\right) = O_\varepsilon(X^{1/2+\varepsilon}).$$

Theorem 4.2 preserves the signed boundary mass, and (11) preserves the power exponent. □

The family may depend on the queried cutoff when the desired conclusion is pointwise and all constants are uniform. A global or online matching claim requires one coherent family.

6. SCOPE AND REMAINING MECHANISMS

Theorem 3.1 rules out square-root performance for short gcd-residual pairs with a fixed or subthreshold edit budget. Theorems 4.2 and 5.2 transfer the same limitation to common-core unit cells under a sparse-jump endpoint budget.

The following mechanisms remain outside the results:

- (i) weighted or signed flows in place of raw unit Möbius mass;
- (ii) overlapping cells with a proved ownership or conservation law;
- (iii) cells without a common gcd or with aggregate residual control;
- (iv) cross-cofactor moves and cancellation among unmatched signs;
- (v) dense-jump budgets with an independent complexity theorem; and
- (vi) analytic estimates not mediated by a short local matching.

Unmatched cardinality can exceed $|M(X)|$ because the unmatched signs may cancel. The conclusions apply to the stated cancellation certificates.

APPENDIX A. VERIFICATION REMARKS

The proofs are symbolic. The threshold theorem combines the displacement factorization (2), the band-product estimate (6), the scale conversion (7), and the strict exponent separation in (9). Exact pair compression preserves $|S_H(X)|$ at every prefix. The sparse-jump argument charges a deleted pair only after establishing $R(m) < R(T_H)$. Finite exact-arithmetic checks recover the displayed cell and its two pairings. The printed deductions establish the asymptotic theorem.

DATA AND CODE AVAILABILITY

The asymptotic arguments are self-contained and use no external dataset. An accompanying Lean 4 development checks finite opposite-sign cancellation, the matching bound, zero-sum cell balance, the strict exponent contradiction, the square-root constant, and the endpoint-doubling connector. The prime-number asymptotics, logarithmic scale conversion, residual gcd model, exact rank compression, and global sparse-jump argument remain manuscript-level proofs.

ASSISTANCE DISCLOSURE

AI systems were used extensively in mathematical derivation, Lean proof development, literature discovery, organization, and typesetting. The author remains responsible for every mathematical statement, proof, citation, and submission decision.

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